

	OPERATING SYSTEMS AND SYSTEM ADMINISTRATION	L	T	P	C
		3	2	0	4

(For Students admitted from 2023 onwards)

Common for B.Sc. (CS) / BCA

COURSE OBJECTIVES:

- To impart knowledge on different types of Operating System
- To learn and implement the concept of Process Management
- To observe the concept of storage management
- To understand the concept of I/O and file systems
- To learn the basics of Linux Programming

COURSE OUTCOMES:

- Understand the fundamental components of a computer operating system.
- Apply the concepts of CPU scheduling, synchronization and deadlocks in real computing problems.
- Demonstrate the different memory and I/O management techniques used in Operating Systems.
- Design and construct the Memory management systems in the modern operating system.
- Have practical exposure to the concepts of Windows, Linux and shell Programming

SYLLABUS

UNIT – I INTRODUCTION

Introduction - Operating system structures - System components - OS services - System calls - System structure -Resources Processes - Threads - Device management.

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UNIT – II PROCESS MANAGEMENT

Processes - Process concepts - Process scheduling - Operations on processes - Cooperating processes - CPU scheduling -Basic concepts -Scheduling criteria - Scheduling algorithms.

12

UNIT – III SYNCHRONIZATION AND DEADLOCKS

The critical section problem - Semaphores - Classic problems of synchronization - Critical regions - Monitors-Dead locks -Deadlock characterization -Prevention - Avoidance - Detection - Recovery.

12

UNIT – IV MEMORY MANAGEMENT

Storage Management Strategies - Contiguous Vs. Non-Contiguous Storage Allocation - Fixed and Variable Partition Multiprogramming - Paging -Segmentation - Paging/Segmentation Systems - Page Replacement Strategies.

12

UNIT – V WINDOWS, LINUX ADMINISTRATION

LINUX Security – LINUX Protection System – LINUX Authorization – LINUX Security Analysis – LINUX Vulnerabilities – Windows Security – Windows Protection System – Windows Authorization – Windows Vulnerabilities.

12

TOTAL: 60

TEXT BOOKS:

1. Abraham Silberschatz, Peter Galvin and Gagne, “Operating System Concepts”, Addison Wesley, 10th Edition, 2018.
2. Harvey M. Deitel,” Operating System”, Addison Wesley, 3rd Edition, 2004.

REFERENCES:

1. Gary Nutt,” Operating System, A modern perspective”, Addison Wesley, 3rd Edition, 2004.
2. Richard Peterson, “Linux: The Complete Reference”, Tata McGraw Hills, 6th Edition, 2008.

	OPERATING SYSTEMS AND SYSTEM ADMINISTRATION LAB	L	T	P	C
		0	0	4	4

(For Students admitted from 2026 onwards)

COURSE OBJECTIVES:

1. To provide an understanding of the design aspects of operating system concepts through simulation
2. Introduce basic Linux commands, system call interface for process management, inter-process communication and I/O in Unix.
3. Student will learn various process and CPU scheduling Algorithms through simulation programs
4. Student will have exposure to System calls and simulate them.
5. Student will learn deadlocks and process management & Inter Process communication and simulate

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LIST OF EXERCISES

- 1 File handling utilities ,Process utilities,Disk utilities Networking commands,Filters,Text processing Utilities Backup utilities 1-7
- 2 Write a shell script that receives any number of file names as arguments checks if every argument supplied is a file or directory and reports accordingly. Whenever the argument is a file it reports no of lines present in it. 8
- 3 Simulate the following CPU scheduling algorithms. a)FCFS b) SJF c) Round Robin d) Priority. 9-14
- 4 Simulate Bankers Algorithm for Dead Lock Avoidance 15-18 Simulate Bankers Algorithm for Dead Lock
- 5 Write a C program to simulate the concept of Dining 19-23 Write a C program to simulate producer
- 6 Write a C program to implement kill(), raise() and sleep()functions. 24-28 Write a C program to implement alarm(), pause() and abort()functions Write a program that illustrate communication between two process using unnamed pipes
- 7 Write a program that illustrates communication between two process using named pipes or FIFO.
- 8 Write a C program that illustrates two processes communicating using Shared memory. 35-36
- 9 Simulate all page replacement algorithms a) FIFO b) LRU c) OPTIMAL 37-42
- 10 Write a C program that takes one or more file/directory names as command line input and reports following information A) File Type B) Number Of Links C) Time of last Access D) Read, write and execute permissions
- 11 a) Implement in c language the following UNIX commands using system calls i) cat ii) ls iii) Scanning Directories (Ex: opendir(),readdir(),etc.)
- 12 Write a C program to simulate disk scheduling algorithms. a) FCFS b) SCAN c) C-SCAN